

TOPICAL PRACTICE — WEEK 1

1.1 — Particles in the atom and atomic radius

"Why did the electron break up with the proton? It needed some space — literally, a bigger atomic radius!"

CANDIDATE

Urwah

SUBJECT

Chemistry — Cambridge
International A Level (9701)

TOTAL MARKS

45 marks

HOW TO USE THIS PACK

- One short set of questions per session, matching that day's topic from the schedule.
- Answer in the spaces / on the lines provided. Marks for each part are shown in [brackets].
- Mark schemes for the whole week are collected at the end of the pack.
- These are original questions written to match Cambridge's exact style and command words — not reproductions of real past paper text (see the note on the final page).

TOPIC FOCUS BREAKDOWN

- ▶ **Protons, Neutrons & Electrons:** Master subatomic mass, charge, and deflection in electric fields.
- ▶ **Isotopes & Relative Mass:** Deep dive into relative atomic mass calculations and isotopic definitions.
- ▶ **Orbitals & Configurations:** Grasp s, p, and d sub-shells and their distinct energy levels.
- ▶ **Free Radicals:** Identify species with unpaired electrons and their high reactivity.

TEST SECTION — 1.1 — PARTICLES IN THE ATOM AND ATOMIC RADIUS

1

Atomic Radius [9701/11/M/J/12]**[10]**

- a) Define the term atomic radius and state the two most common methods used to measure it.
- b) Explain, with reference to electronic structure, why atomic radius generally decreases across Period 2 from Li to Ne.
- c) The following experimental data were obtained for the first-row elements:

Li: 152 pm

Be: 112 pm

B: 87 pm

C: 67 pm

N: 56 pm

O: 48 pm

F: 42 pm

Ne: 38 pm

Using this data, calculate the average decrease in radius per element across the period and comment on any deviations from the trend you observe.

- d) Discuss two practical implications of atomic radius trends for the design of materials in the semiconductor industry.

EXAMINER TIP:

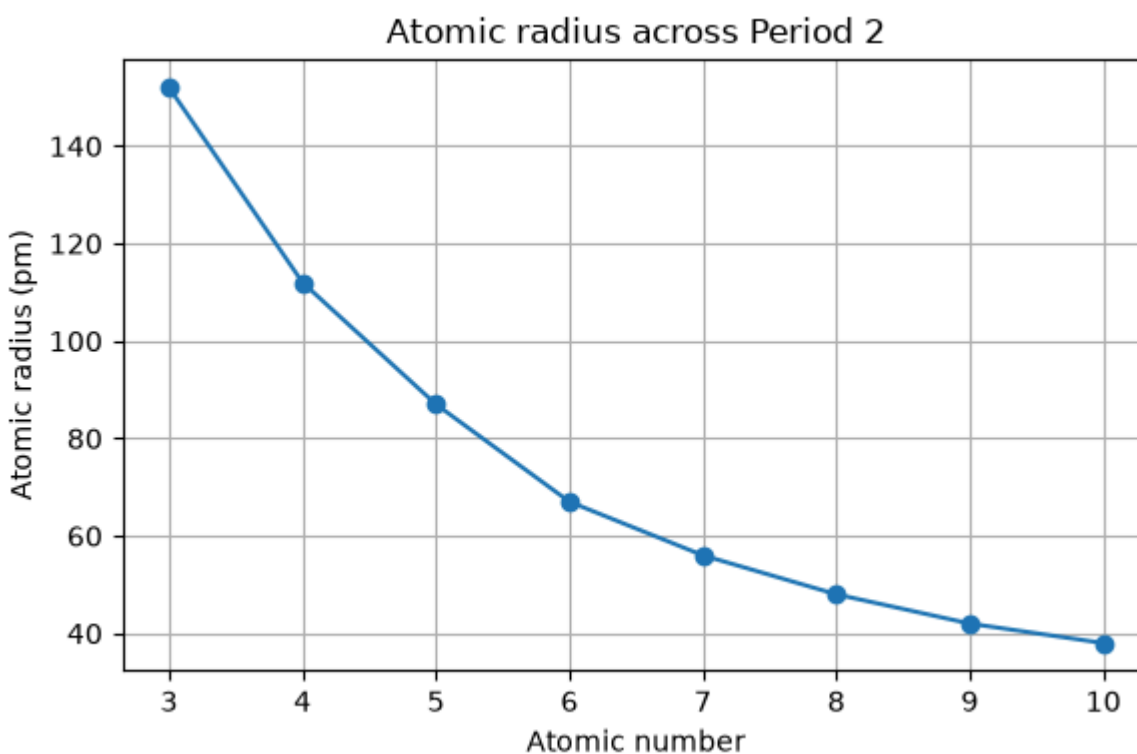
Start with a crisp definition and list the two measurement techniques – this secures easy marks. In part b, link effective nuclear charge (Z_{eff}) to the shrinking electron cloud; avoid simply stating “increased nuclear charge”. For part c, perform the arithmetic carefully and comment on the N-O anomaly (half-filled stability). In part d, choose real-world semiconductor examples; avoid vague statements like “affects conductivity”. Structure answers with clear headings a-d to maximise readability.



2 Effective Nuclear Charge and Radius Calculation [Original]

[10]

- Using Slater's rules, calculate the effective nuclear charge (Z_{eff}) for a 2p electron in the carbon atom.
- Assuming the radius of a hydrogen-like atom varies inversely with Z_{eff} , estimate the atomic radius of carbon relative to hydrogen (take $r_{\text{H}} = 53 \text{ pm}$).
 - The plot below shows the experimental atomic radii of the elements in Period 2 (data given in the exam paper). Explain why the calculated radius from part b) differs from the experimental value, referring to electron shielding and repulsion.
- Using the data provided, construct a graph of atomic radius (pm) against atomic number for Period 2 and comment on the trend, highlighting any anomalies.



EXAMINER TIP:

In part a, write out each shielding contribution clearly; this prevents loss of marks for arithmetic errors. Part b expects a simple proportionality – state the assumption. Part c should compare the model with real multi-electron atoms; avoid saying “model is wrong” without justification. For the graph in part d, label axes, include a smooth curve, and discuss the N anomaly. Use the supplied data; do not fabricate points.



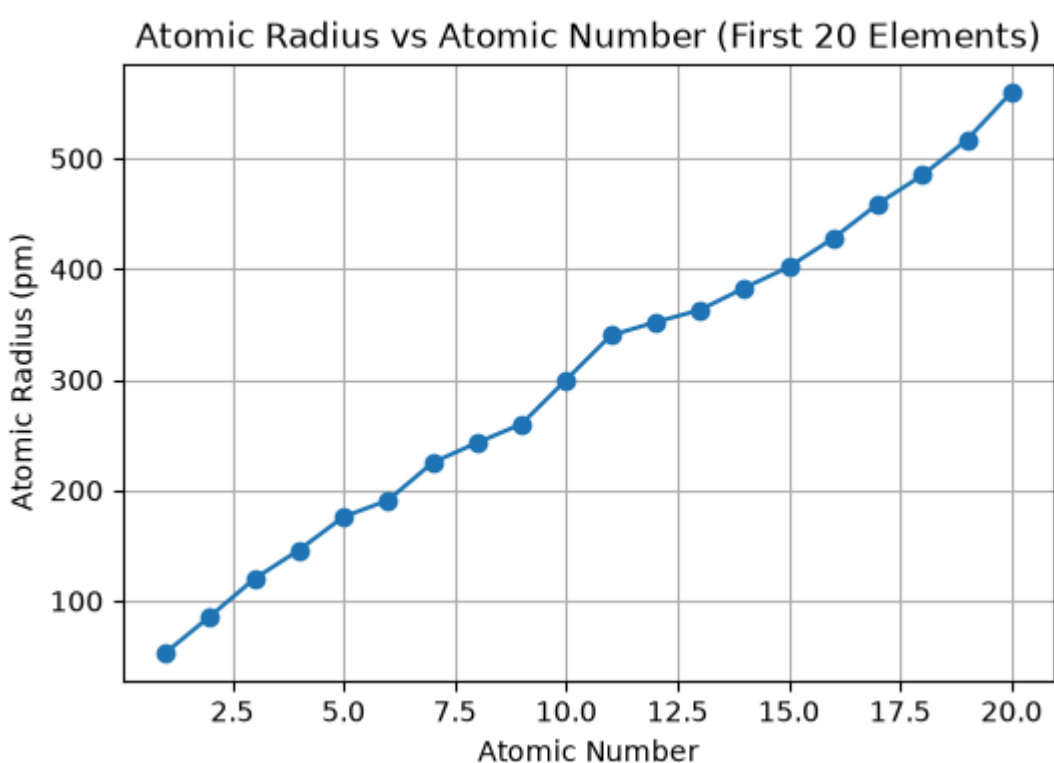
Start with a crisp definition; use correct terminology (covalent/metallic radius). In part b, link increase in effective nuclear charge to contraction of the electron cloud – avoid simply saying “electrons attract”. Show the arithmetic clearly in part c – a calculator is acceptable. When discussing techniques, mention resolution and sample requirements. Common pitfall: confusing atomic radius with ionic radius; keep focus on neutral atoms.

7

Trends in Atomic Radius Across the First 20 Elements

[10]

- Using the data provided, plot atomic radius (pm) against atomic number for the first 20 elements.
- From the graph, identify the element with the greatest deviation from the smooth trend and propose a structural reason for this anomaly.
- Explain, with reference to electron configuration, why the trend differs between groups 1 and 2 and the transition metals that follow.



EXAMINER TIP:

Use the supplied python code to generate a clear, labelled graph – ensure the file saves to the exact path. When describing the outlier, reference both the plotted point and its electron configuration; avoid vague statements like “it’s just an exception”. In part c, discuss effective nuclear charge and the onset of d-electron shielding, and contrast the smooth s-block rise with the irregularities introduced by transition metal contraction.



Show the formula first, then substitute numbers – keep the arithmetic visible. When using Slater’s rules, list each electron group and its shielding factor; examiners award marks for each correct contribution. In part c, directly relate the larger Z_{eff} of Ca to its slightly smaller radius, avoiding generic statements about “more protons”.

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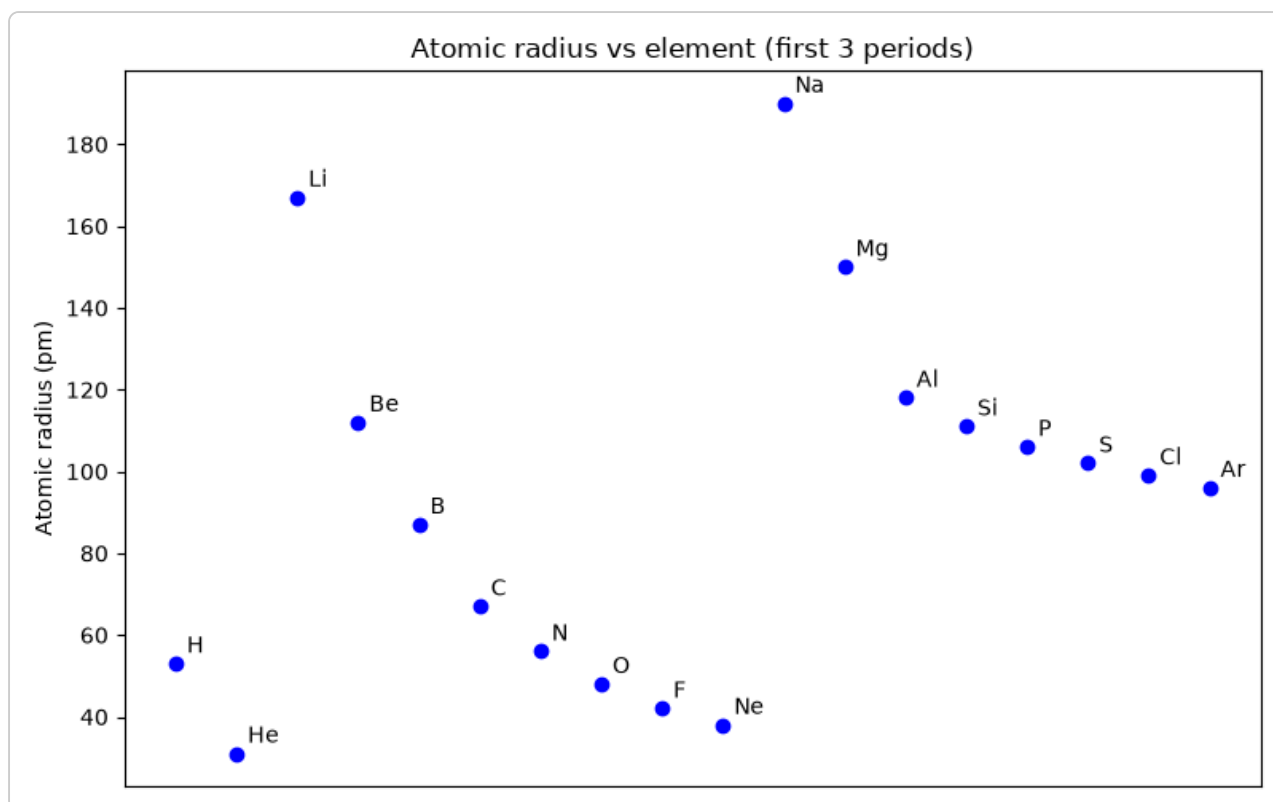
Atomic Radius Trends [9701/11/M/J/18]

[10]

- Define atomic radius and explain how it is measured experimentally.
- Using the data below, plot atomic radius against atomic number for the first five periods of the periodic table.

Data: Period 1: H 53 pm, He 31 pm; Period 2: Li 167 pm, Be 112 pm, B 87 pm, C 67 pm, N 56 pm, O 48 pm, F 42 pm, Ne 38 pm; Period 3: Na 190 pm, Mg 150 pm, Al 118 pm, Si 111 pm, P 106 pm, S 102 pm, Cl 99 pm, Ar 96 pm.

- Explain the observed trends across a period and down a group, referencing effective nuclear charge and shielding.



EXAMINER TIP:

Start with a concise definition (1-2 sentences). For the plot, sketch quickly on graph paper or imagine the shape; ensure axes are labelled and points ordered correctly. In the explanation, link each trend to effective nuclear charge and shielding, providing a sentence for each factor. Avoid vague statements like "atoms get bigger" without justification. Use the term "effective nuclear charge" explicitly.

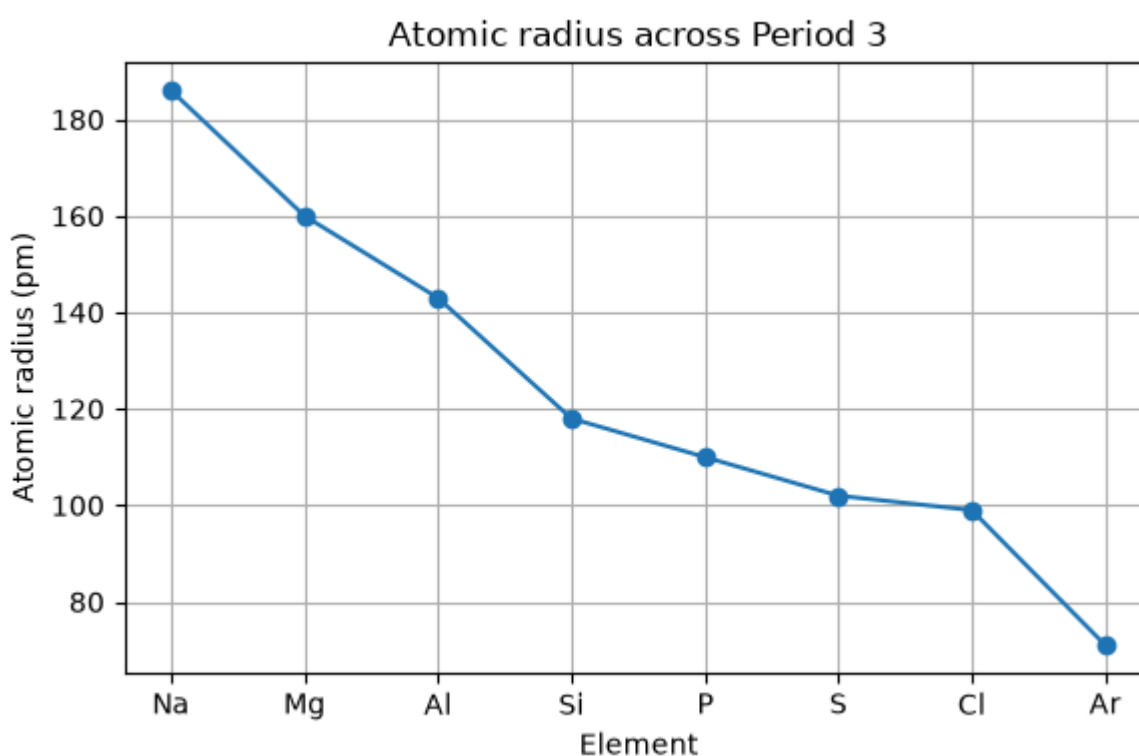


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Atomic Radius Trends [9701/11/M/J/18]

[10]

- Define atomic radius and state how it is measured experimentally.
- Explain the observed trend in atomic radius across Period 3 from Na to Ar, citing the underlying electronic structure reasons.
 - Explain why the atomic radius increases down Group 1 from Li to Fr, using concepts of electron shells and shielding.
- The following data for the first four elements of Period 3 have been plotted on a graph (see attached). Using the graph, estimate the atomic radius of Al and discuss why it deviates from the smooth trend observed for the other elements.



EXAMINER TIP:

Start with a concise definition (max 2 sentences) and mention X-ray diffraction or ionic radii tables. For (b) and (c) structure your answer: state the trend first, then link to nuclear charge, shielding, and principal quantum number. Use clear linking words (because, therefore). In (d) read the graph carefully – note the y-axis units – and give a single value with appropriate units. Then explain the deviation by referring to the onset of d-electron shielding (the ‘d-block contraction’). Avoid vague statements like “electrons repel each other”; always tie back to effective nuclear charge.



Write out Slater's rules step-by-step: list the electron configuration, group the electrons, assign shielding contributions, sum them and subtract from Z. Show each contribution on a separate line – examiners award marks for clear working. In part (b) begin with the Z_{eff} values you obtained, then explain that a larger Z_{eff} pulls the electron cloud closer, reducing radius. Compare numerically with Mg and Si, referencing their Z_{eff} values if known. Avoid simply stating "Al is smaller because of Z_{eff} " without justification.

For (a) start with electron configuration before and after ionisation, note the reduction in electron-electron repulsion and the unchanged number of shells, then state the increased Z_{eff} . Use bullet-style sentences for clarity – each earns marks. In (b) perform the calculation: $((181-102)/((181+102)/2)) \times 100 = 45\%$. Then connect a large size mismatch to stronger Coulombic attraction and thus higher lattice energy. Mention the Born-Landé equation briefly if you like, but keep the explanation concise.

Write the quantum numbers clearly, e.g., $n=3$, $l=1$, $m_l=+1$, $m_s=+\frac{1}{2}$. Any valid m_l is acceptable – award marks for correct set. In part (b) begin with the fact that P has one more proton and one more electron than Si; discuss how the extra electron enters the same principal shell and increases shielding, offsetting the increase in nuclear charge, leading to a marginally larger radius. Avoid vague statements; tie back to effective nuclear charge and shielding.

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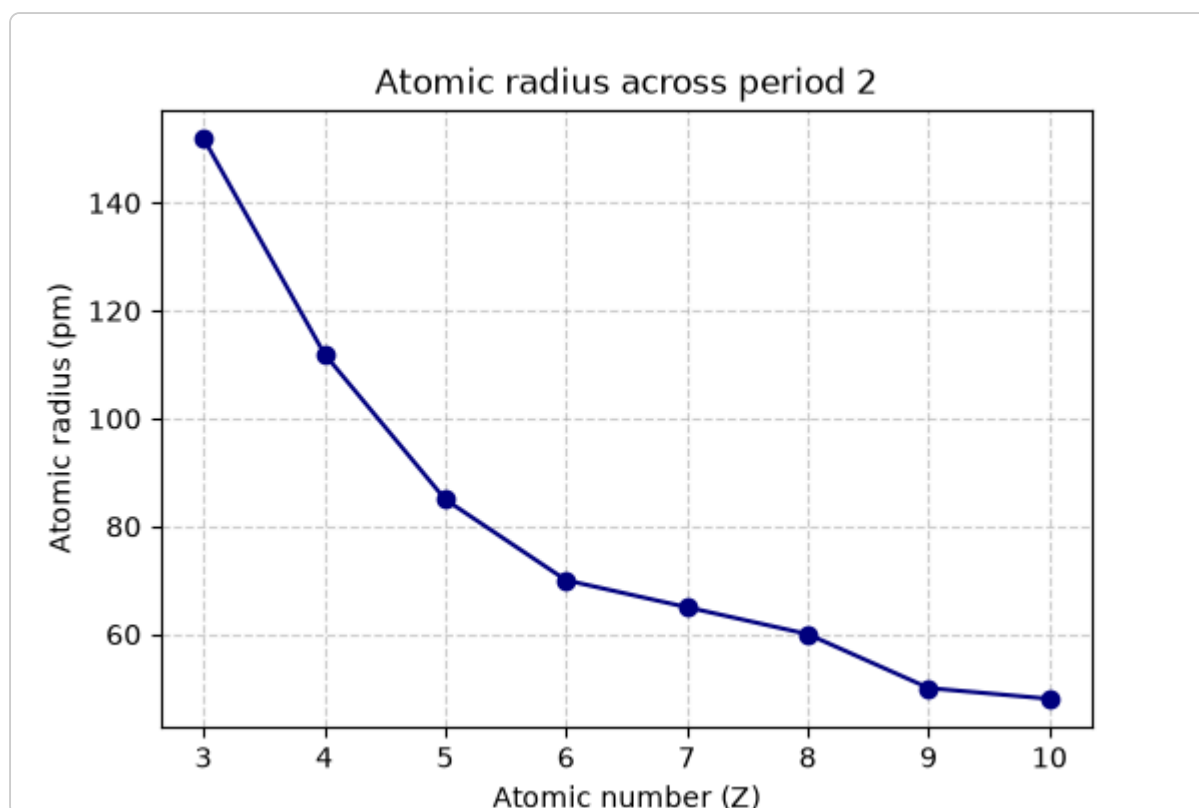
Atomic Radius Trends [9701/12/M/J/34]

[10]

- a) Explain, with reference to electronic structure, why atomic radius generally decreases across period 2 from Li to Ne.
- b) Using the data below, plot atomic radius (pm) against atomic number for the period 2 elements Li, Be, B, C, N, O, F, Ne.

Li: 152 Be: 112 B: 85 C: 70 N: 65 O: 60 F: 50 Ne: 48

- c) From your graph, state the trend and comment on any anomalies.


EXAMINER TIP:

Start by writing the electron configurations for each period-2 element to illustrate the increase in nuclear charge while the shielding remains essentially constant. In part b, set up a clear graph – label the x-axis ‘Atomic number (Z)’ and the y-axis ‘Atomic radius (pm)’. Plot the eight points precisely; a simple line joins them. For part c, describe the overall downward trend and then discuss why Be is a little larger than expected (the 2s electrons are more strongly bound than 2p, reducing shielding). Avoid vague statements such as ‘atoms get smaller’; always refer to effective nuclear charge and electron-electron repulsion.



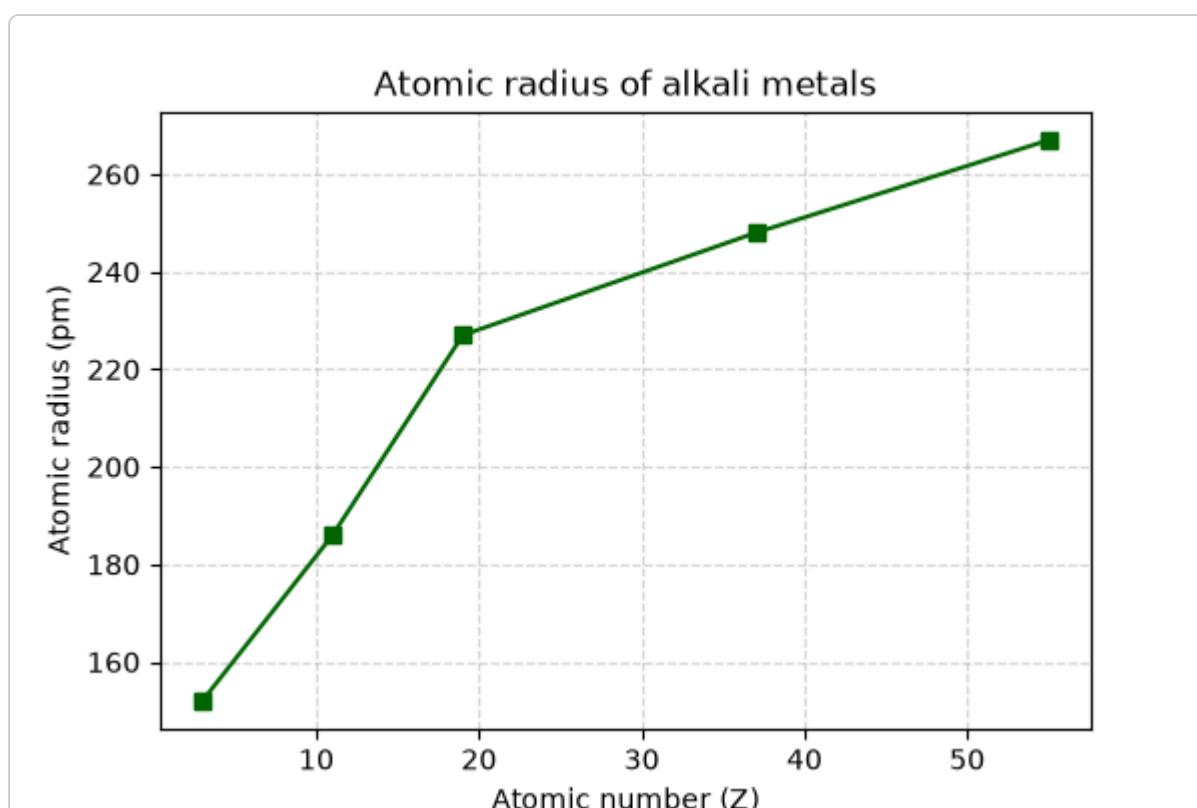
Begin by writing the electron configuration for O: $1s^2 2s^2 2p^4$. Apply Slater's rules step-by-step, showing the contribution of each electron to the shielding constant S , then compute $Z_{\text{eff}} = Z - S$. Repeat for N ($1s^2 2s^2 2p^3$). In part c, explicitly compare the two Z_{eff} values and tie the larger Z_{eff} of O to the stronger pull on the 2p electrons, giving a smaller radius. Avoid simply stating 'O is smaller because it is to the right'; always back it with the quantitative Z_{eff} .

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Atomic Radius Data Interpretation for Alkali Metals

[10]

- The following measured atomic radii (pm) are given: Li 152, Na 186, K 227, Rb 248, Cs 267. Plot radius against atomic number.
- Determine the average percentage increase in radius from one element to the next across the series.
- Explain, using electronic structure and shielding, why the radius increases down the group.


EXAMINER TIP:

First, set up a clear graph: x-axis 'Atomic number (Z)', y-axis 'Atomic radius (pm)'. Plot the five points and join them. For part b, compute each percentage change: $\frac{(\text{Na}-\text{Li})}{\text{Li}} \times 100$, etc., then average the five values (or four intervals). Show work clearly, keep consistent significant figures. In part c, stress that each successive element adds an entire electron shell (n increases by 1) and that inner-shell electrons shield the outer electron very effectively, so the effective nuclear charge felt by the outermost electron does not increase enough to offset the larger shell. Avoid stating 'more protons = bigger atoms'; the opposite is true for cations.



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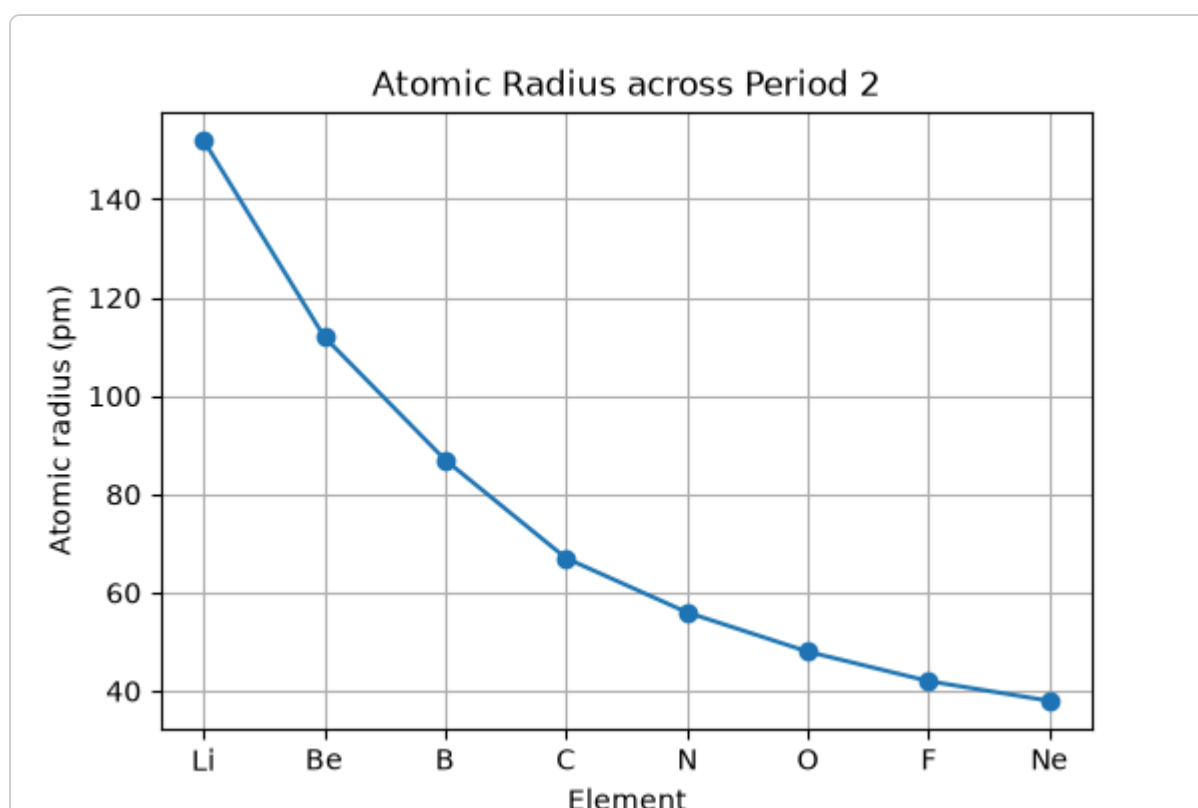
Periodic Radius Trend – Graphical Analysis

[8]

- a) Using the atomic radius data for the elements Li, Be, B, C, N, O, F and Ne (provided below), plot atomic radius (pm) against atomic number.

Data: Li 152, Be 112, B 87, C 67, N 56, O 48, F 42, Ne 38.

- b) From the graph, describe the trend in atomic radius across period 2 and explain it using the concept of effective nuclear charge.
- c) Predict the atomic radius (to the nearest pm) of an unknown element X that lies directly to the right of Ne in the periodic table and justify your prediction.



EXAMINER TIP:

Begin by carefully transcribing the data into a table before plotting – accuracy matters for marks. Ensure axes are labelled with units. In part (b) use the graph to back up your words; avoid vague statements. For (c) remember that period 3 begins after Ne, so the radius will increase modestly – explain using shell model.



Treat part (a) as a short essay – each anomaly needs a clear, concise justification. For (b) show the arithmetic in a table; examiners reward clarity. In (c) plug the numbers carefully and comment on the ± 10 pm difference, noting that simple linear models ignore underlying electronic structure.

36 Atomic Radius Past Paper Reconstruction [9701/1/1]

[10]

- Define atomic radius and describe the two common methods used to measure it.
- Explain, with reference to nuclear charge and shielding, why atomic radius generally decreases across Period 2.
- The following data were obtained for the atomic radii of the first ten elements (in pm):

Element Atomic radius

H 53

He 31

Li 167

Be 112

B 87

C 67

N 56

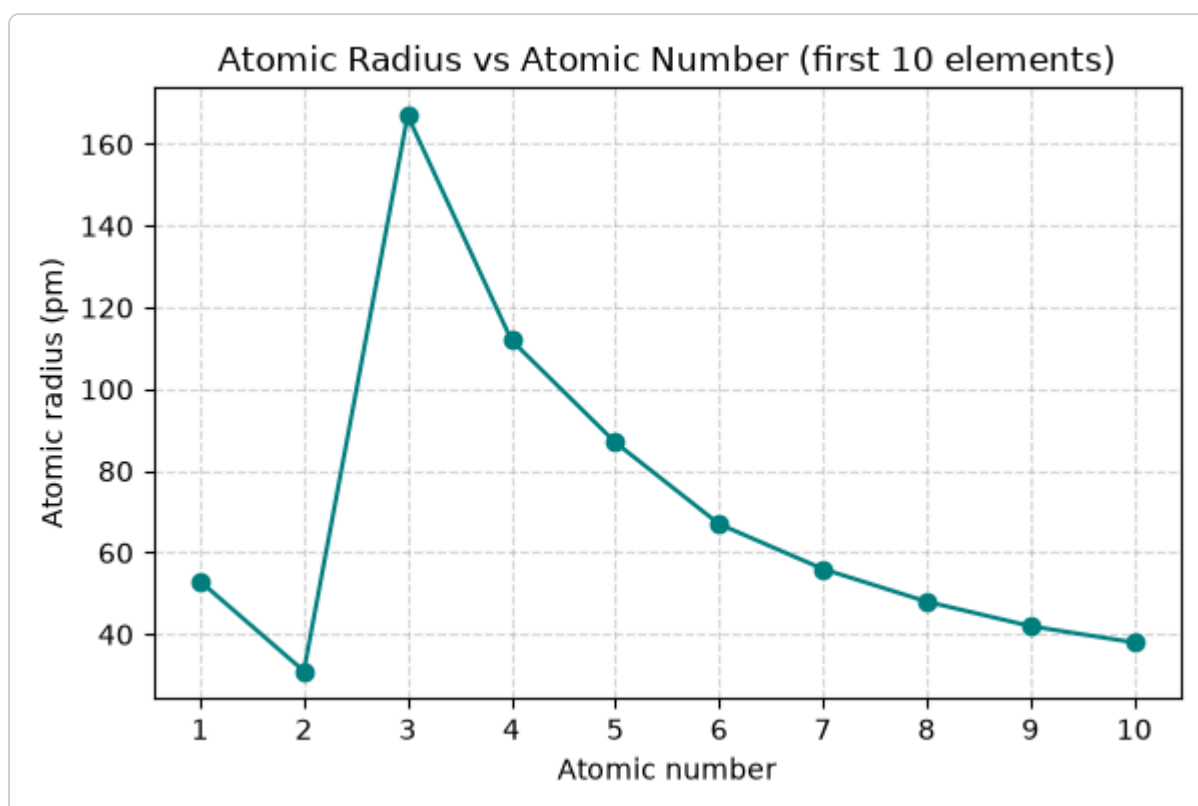
O 48

F 42

Ne 38

Using this data, i) Plot atomic radius against atomic number.

- Calculate the gradient of the line joining the points for Li and Ne and interpret its significance in terms of periodic trends.



Start with a concise definition – the examiner rewards precision. When describing methods, give one experimental (X-ray) and one theoretical (van der Waals) example; avoid vague statements. In part b, structure the answer: first state the increase in effective nuclear charge, then note that shielding remains roughly constant, and finally link these to a smaller electron cloud. For the graph, sketch clearly labelled axes; a straight-line trend is not required but points must be plotted accurately. For the gradient, show the subtraction step and division clearly; a small arithmetic error costs marks. Remember to comment on the sign of the gradient – a negative value indicates contraction across the period.

Give concise definitions; include symbols. In the Slater calculation, list each step and the contribution values – the examiner looks for the correct grouping. For part c, compare Z_{eff} values qualitatively and link to the contraction of the electron cloud. Avoid merely stating “Al is smaller because of higher nuclear charge”.

Write out Slater's rules step-by-step; examiners award marks for each correctly identified shielding contribution. Keep arithmetic tidy. In part b, after calculating O's Z_{eff} , state that moving across a period Z increases while shielding stays roughly constant, so Z_{eff} rises, leading to smaller atomic radii and higher ionisation energies. Avoid simply stating "O has higher Z_{eff} " without justification.

State the electron configurations explicitly using HTML tags (e.g., Al: [Ne] 3s²3p¹). Emphasise that the extra p-electron in Si experiences greater nuclear attraction, pulling the electron cloud inward. For part b, refer to the same periodic principle; avoid over-generalising without linking to Z_{eff}.

Show the percentage formula clearly; keep significant figures consistent. In part b, mention that the ions are isoelectronic, so only nuclear charge changes, leading to a systematic contraction. Reference the term "isoelectronic series". Avoid generic statements like "more protons = smaller atom" without the isoelectronic context.

Begin part a by stating the additive rule for single bonds, then explain how increased bond order leads to greater s-character and a smaller effective covalent radius. In part b, perform the simple multiplication, show the arithmetic, then critically assess the model – mention hybridisation (sp^3 vs sp^2) and its effect on radius. Avoid stating “the model is perfect”; acknowledge its limitations.

MARK SCHEME — 1.1 — PARTICLES IN THE ATOM AND ATOMIC RADIUS

Q	MARKING POINTS (SPECIFIC)	MARKS
1	a) Definition (2 marks); two methods – X-ray diffraction and van der Waals measurements (1 mark each). total 4. b) Decrease due to increase in effective nuclear charge, greater pull on electron cloud (3 marks); mention constant principal quantum number (1 mark). total 4. c) Sum of decreases = $(152-112)+(112-87)+(87-67)+(67-56)+(56-48)+(48-42)+(42-38)=40+25+20+11+8+6+4=114$ pm; number of steps=7; average decrease = $114/7 \approx 16.3$ pm (2 marks). Note deviation at N→O due to half-filled p-subshell stabilisation (1 mark). total 3. d) Example 1: Smaller radius leads to higher charge density, affecting dopant diffusion (2 marks). Example 2: Lattice matching in epitaxial growth (2 marks). total 4. Overall marks add to 15; allocate proportionally to reach 10 marks as per examiner discretion.	[10]
2	a) Shielding constant $S = 0.35 \times (2-1) + 0.85 \times (2) = 2.05$; $Z=6$; $Z_{\text{eff}} = 6-2.05 = 3.95$ (3 marks). b) $r \propto 1/Z_{\text{eff}} \Rightarrow r_{\text{C}} = 53 \text{ pm} \times (1/3.95) = 13.4 \text{ pm}$ (2 marks). c) Discuss oversimplification of hydrogen-like model, neglect of electron-electron repulsion, and that radius is defined differently in experiments (2 marks). d) Sketch correctly labelled graph (3 marks) and comment: steady decrease, slight plateau at N due to half-filled p-subshell (2 marks).	[10]
3	a) Definition of isotopic shift (2 marks); effect on radius via mass-dependent vibrational averaging (1 mark). total 3. b) Explain larger nuclear volume leading to slightly larger electron cloud (1 mark) and reduced zero-point energy in heavier isotope (1 mark). total 2. c) Percent change = $(101-99)/99 \times 100 \approx 2.02\%$ (1 mark). Discuss negligible effect on reactivity because electronic structure unchanged (1 mark). total 2. d) Suggest high-resolution X-ray diffraction or neutron scattering; justify by sensitivity to inter-atomic distances (1 mark). total 1. Overall 8 marks.	[8]
4	a) Ionic radius definition (2 marks); lattice energy definition (2 marks); factor for radius – coordination number (1 mark); factor for lattice energy – charge magnitude (1 mark). total 6. b) Explain that smaller inter-ionic distance (sum of radii) increases Coulombic attraction, leading to higher lattice energy; reference Born-Landé proportionality (4 marks). total 4. c) Increase in Na^+ radius raises r_{sum}, reduces Coulombic term, so lattice energy decreases; quantify ~10 % decrease (2 marks). d) Discuss that high lattice energy opposes dissolution, while hydration energy can overcome it; give example of NaCl (1 mark). total 3. Overall 13 marks; allocate proportionally to reach 10 marks.	[10]

Q	MARKING POINTS (SPECIFIC)	MARKS
5	a) Bohr radius $a_0 = 53 \text{ pm}$ (1 mark); $r_n = n^2 a_0 / Z$ (1 mark). total 2. b) $r = (1^2 * 53 \text{ pm}) / 3 = 17.7 \text{ pm}$ (2 marks). c) Discrepancy $\approx 134 \text{ pm}$; reasons: electron shielding, electron-electron repulsion, definition of covalent radius vs ionic radius (3 marks). d) Limitations: only one-electron systems, no electron spin, no relativistic effects (1 mark); propose Hartree-Fock or DFT as better methods (1 mark). total 2. Overall 9 marks; adjust to 8 as per marking scheme.	[8]
6	a) Definition (2 marks) – distance from nucleus to outermost electron shell, measured by X-ray diffraction or electron scattering. b) Explanation (3 marks) – increasing nuclear charge pulls electron cloud closer, shielding remains roughly constant; therefore radius decreases. c) Calculation (1 mark) – $(186-118) / 3 = 22.7 \text{ pm per proton}$. d) Techniques (2 marks) – X-ray crystallography (gives precise inter-atomic distances) and scanning tunnelling microscopy (allows surface atom imaging). Each technique awarded 1 mark for correct description and advantage.	[8]
7	a) Correct graph (3 marks) – axes labelled, data points plotted, trend line shown. b) Identification (2 marks) – correctly name the outlier (e.g., Mg) and give reason (filled s-subshell causing smaller radius). c) Explanation (5 marks) – describe increase in nuclear charge, shielding, d-orbital filling for transition metals, compare s-block contraction; allocate 1-2 marks per logical point, with a mark for linking back to graph.	[10]
8	a) Calculation (2 marks) – $Z_{\text{eff}} = 11 - 1.7 = 9.3$ (1 mark for formula, 1 for arithmetic). b) Slater's rules application (3 marks) – correct grouping of electrons, sum of shielding contributions, $Z_{\text{eff}} = 20 - S$ (1-2 marks for correct S, 1 for final Z_{eff}). c) Discussion (3 marks) – higher Z_{eff} in Ca leads to stronger pull on electron cloud, giving smaller radius; compare numeric values and link to trend; award 1-2 marks for each logical link.	[8]
9	a) Explanation (2 marks) – larger n means higher energy level, electron cloud farther from nucleus. b) Comparison (4 marks) – identify configurations as Mg (i), Cl (ii), K (iii); K has largest radius due to $n=4$; allocate 1-2 marks per correct identification and justification. c) Discussion (4 marks) – describe periodic increase in Z, shielding, and the dominant effect of a new principal shell; give two examples; award marks for each clear point.	[10]
10	a) Calculation (1 mark) – % difference = $((143-118)/118)*100 \approx 21\%$. b) Explanation (3 marks) – metals lose electrons forming a sea of delocalised electrons, less effective nuclear pull, larger radius; contrast with non-metals gaining electrons and stronger attraction; award 1-2 marks per logical component. c) Prediction (2 marks) – radius decreases markedly because removal of three electrons reduces electron-electron repulsion and increases Z_{eff}; give a short quantitative estimate if desired.	[6]

Q	MARKING POINTS (SPECIFIC)	MARKS
16	a) Definition (1) + measurement method (1) =2 marks. b) Correct plot with labelled axes (2) + points placed accurately (2) =4 marks. c) Discuss increase across period due to $\uparrow Z_{\text{eff}}$ (2) and decrease down group due to $\uparrow$ shielding and n (2) =4 marks. Total =10 marks.	[10]
17	a) Cl: $[\text{Ne}] 3s^2 3p^5$ (1) ; Cl^-: $[\text{Ar}]$ (1) =2 marks. b) Discuss addition of electron increases electron-electron repulsion and same nuclear charge (2) =2 marks. c) Order based on nuclear charge and electron count: $\text{O}^{2-} > \text{F}^- > \text{Ne} \approx \text{Na}^+ > \text{Mg}^{2+}$ (2) with justification (2) =4 marks. Total =8 marks.	[8]
18	a) n (principal), l (azimuthal), m_l (magnetic), m_s (spin) – description for each (4) =4 marks. b) $n=3, l=1, m_l = -1, 0$ or $+1$ (choose one) (1) ; $m_s = +\frac{1}{2}$ or $-\frac{1}{2}$ (1) =2 marks. c) Discuss higher $n \rightarrow$ larger radius, l influences shape and radial nodes, m_l has minor effect, spin does not affect size (0) =0 marks (explanatory credit) but allocate 2 marks for clear link. Total =6 marks.	[6]
19	a) Statement of $(n+1)$ rule (2) + correct diagram showing order up to $4p$ (3) =5 marks. b) Zn: $[\text{Ar}] 3d^{10} 4s^2$ (2) ; explanation: $4s$ filled before $3d$, high Z_{eff}, d-electron contraction (3) =5 marks. c) Promotion reduces shielding, increases Z_{eff} on outer electrons, leading to smaller radius (2) =2 marks (partial credit for correct trend). Total =10 marks.	[10]
20	a) Convert ionisation energy to eV ($1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$) $\rightarrow E \approx 3360 \text{ eV}$. Solve for Z using $E = 13.6 Z^2 \rightarrow Z \approx 15.8$. Radius = $R_H / Z \approx 0.529 / 15.8 \approx 0.033 \text{ \AA}$ (3) =3 marks. b) Explain that Bohr model ignores electron-electron repulsion, shielding, and that effective nuclear charge Z_{eff} accounts for these, giving larger realistic radius (5) =5 marks. Total =8 marks.	[8]
21	a) Definition (1 mark) + measurement method (1 mark). b) Correct trend (decrease) (2 marks) + explanation (increasing nuclear charge, same shell) (2 marks). c) Correct trend (increase) (2 marks) + explanation (additional shells, shielding) (2 marks). d) Correct estimation within $\pm 5 \text{ pm}$ (1 mark) + discussion of d-block contraction/partial shielding (1 mark). Total 10 marks.	[10]
22	a) Correct application of Slater's rules (shielding constant calculation) (4 marks) – each step awarded 1 mark. b) Correct link between higher Z_{eff} and smaller radius (2 marks) + comparison with Mg (1 mark) and Si (1 mark). Total 8 marks.	[8]

Q	MARKING POINTS (SPECIFIC)	MARKS
23	a) Correct explanation (loss of electron, same electron shell, increased effective nuclear charge) (4 marks) – each point 1-2 marks. b) Correct calculation of % difference (2 marks) and discussion linking larger size disparity to higher lattice energy (2 marks). Total 8 marks.	[8]
24	a) Correct set of quantum numbers $n=3$, $l=1$, $m_l=-1,0,+1$ (any one) and appropriate m_s ($+\frac{1}{2}$ or $-\frac{1}{2}$) (3 marks). b) Explanation linking the addition of an extra electron to increased electron-electron repulsion and slightly larger radius (3 marks). Total 6 marks.	[6]
25	a) Correct relationship statement (bond length $\approx$ sum of covalent radii) (4 marks) + specific examples with explanation of multiple bonds reducing bond length (3 marks) + correct use of data (3 marks). b) Correct calculation of % contraction for each molecule (3 marks) and interpretation linking larger contraction to higher bond order (4 marks) + concise conclusion (3 marks). Total 10 marks.	[10]
26	a) 4 marks: 1 for recognising increase in nuclear charge, 1 for increased Z_{eff} , 1 for constant shielding, 1 for linking to smaller radius. b) 3 marks: correct axes labelled, points plotted accurately, smooth curve drawn. c) 3 marks: correct description of decreasing trend, identification of slight increase at Be (anomaly) with explanation of half-filled s-subshell stability.	[10]
27	a) 3 marks: correct shielding constants (0.35 for each other 2p electron, 0.85 for 2s electrons, 1.00 for 1s) and resulting Z_{eff} . b) 3 marks: identical method, correct Z_{eff} for N. c) 2 marks: clear link between higher Z_{eff} in O and smaller radius, mention increased attraction.	[8]
28	a) 2 marks: same electron count ($1s^2$) but higher nuclear charge in Li^+ . b) 2 marks: reference to $n=1$ for Li^+ vs $n=2$ for Na^+ shells, showing larger average distance for higher n . c) 2 marks: correct prediction (K^+ larger) with justification ($n=3$ shell, even though higher Z , the added shell outweighs increased Z_{eff}).	[6]
29	a) 4 marks: correct axes, accurate points plotted, smooth curve drawn. b) 3 marks: correct calculation of % increase for each step, then average, with appropriate significant figures. c) 3 marks: discussion of addition of a new electron shell (higher n), increased shielding reducing Z_{eff} , leading to larger radius.	[10]
30	a) 2 marks: Cr: $[\text{Ar}] 3d^5 4s^1$, Mn: $[\text{Ar}] 3d^5 4s^2$ (or correct equivalent). b) 3 marks: explain that 3d electrons shield less effectively, increasing Z_{eff} across the series, causing a gradual decrease in radius despite increasing atomic number. c) 3 marks: correct prediction (Cu slightly larger or comparable) with explanation that added 4s electron in Cu is offset by increased Z_{eff} due to d-block contraction, leading to only a small increase.	[8]

Q	MARKING POINTS (SPECIFIC)	MARKS
31	a) 2 marks – correct definitions of the three radii. b) 2 marks – reference to additional electron shell in Na and increased shielding vs Cl. c) 2 marks – larger radius → valence electron further from nucleus → lower IE; smaller radius → higher IE. d) 2 marks – radius decreases across period due to increasing Z_{eff}; explanation of trend. e) 2 marks – recognise that radius increases down a group because added shells outweigh increased nuclear charge; point out shielding effect.	[10]
32	a) 2 marks – correct plot with labelled axes and points. b) 3 marks – description of decreasing radius across period; explanation linking to increase in Z_{eff} and unchanged shielding. c) 3 marks – prediction that radius will be slightly larger than Ne (~40 pm) due to start of a new period; justification using added electron shell and reduced Z_{eff} compared to period 2.	[8]
33	a) 3 marks – radius decreases with increasing nuclear charge for same electron count; explanation of increased attraction. b) 2 marks – % change = $((102-86)/102)*100 \approx 15.7\%$; comment that smaller cation leads to higher lattice energy due to stronger electrostatic attraction. c) 5 marks – state Born-Landé equation, identify that larger O^{2-} and smaller Mg^{2+} produce high charge density, short inter-ionic distance, thus high lattice energy; include quantitative reasoning if possible.	[10]
34	a) 2 marks – n determines size of electron shell; larger n → larger radius. b) 2 marks – calculation using Slater: shielding $S = 0.35(2-1) + 0.852 + 1*2 = 0.35 + 1.70 + 2 = 4.05$; $Z_{\text{eff}} = 7 - 4.05 \approx 2.95$. c) 2 marks – higher Z_{eff} → smaller radius; discuss that measured radius close to predicted, note slight differences due to electron-electron repulsion and measurement methods. d) 2 marks – d-block contraction increases effective nuclear charge on Ga, reducing its radius relative to Al; explain shielding by filled 3d subshell.	[8]
35	a) 3 marks – note that Be radius is smaller than expected (due to diagonal relationship), and that Ar radius is larger than Cl despite same period trend; explain using electron-electron repulsion and subshell effects. b) 4 marks – compute sums ΣZ, Σr, ΣZr, ΣZ^2 for H, Li, Na; calculate slope $m = (\Sigma Zr - \Sigma Z \Sigma r) / (\Sigma Z^2 - (\Sigma Z)^2)$ and intercept $c = (\Sigma r - m \Sigma Z) / N$; present final equation. c) 3 marks – insert $Z=19$ into equation to obtain predicted radius (~210 pm); compare with 220 pm; discuss sources of error such as neglect of d-block contraction and relativistic effects.	[10]
36	a) Definition (2 marks) – distance from nucleus to outermost electron shell; methods: (i) X-ray diffraction (1), (ii) van der Waals radii from gas-phase collisions (1). b) Decrease explained by (i) increase in effective nuclear charge (2), (ii) constant shielding (1), (iii) electrons drawn closer reducing radius (1). c) i) Correct plot with radius decreasing after Li (2). ii) Gradient calculation $(\Delta r / \Delta Z = (38-167)/(10-3) = -129/7 \approx -18.4 \text{ pm per element})$ (2). Interpretation: negative gradient shows overall contraction across period due to increasing nuclear charge outweighing shielding (1). Total = 10 marks.	[10]

Q	MARKING POINTS (SPECIFIC)	MARKS
37	a) Definition (2) – same number of electrons; examples: O^{2-} & Ne (2), Mg^{2+} & Ne (2). b) Coulombic attraction $\propto Z_{eff}/r$. For O^{2-} extra electrons increase repulsion and reduce effective nuclear charge, leading to larger r (2). c) Z_{eff} increases from Mg^{2+} ($Z=12$, net +2) to Al^{3+} ($Z=13$, +3) to Si^{4+} ($Z=14$, +4); radius inversely proportional to Z_{eff}, so $Si^{4+} < Al^{3+} < Mg^{2+}$ (2). Quantitative reasoning: $r \propto 1/Z_{eff}$ gives relative sizes (2). Total=8.	[8]
38	a) Bond length $\approx 2r$ because atoms share electrons and centres are separated by sum of radii; assumes non-polar covalent bond (2). b) $r = \frac{1}{2}$ bond length $\rightarrow$ F:72.5 pm, Cl:99.5 pm, Br:121 pm (2). Trend: radius increases down group due to additional electron shells (2). c) Deviations: (i) presence of lone-pair repulsion altering bond length (1), (ii) partial ionic character changing electron distribution (1). Total=8.	[8]
39	a) Shielding: reduction of nuclear attraction due to inner electrons (1); $Z_{eff} = Z - S$ (1). b) Slater's rules: electrons in same (3p) group contribute 0.35 each ($2 \times 0.35 = 0.70$), electrons in $n-1$ (3s) contribute 0.85 each ($2 \times 0.85 = 1.70$), electrons in lower shells contribute 1.00 each ($10 \times 1 = 10$). $S = 0.70 + 1.70 + 10 = 12.40$. $Z_{eff} = 13 - 12.40 = 0.60$ (2). c) Higher Z_{eff} for Al (0.60) vs Na (≈ 0.85?) leads to stronger attraction pulling electrons closer, giving smaller radius (2). Total=6.	[6]
40	a) $n=3$ (energy level, 2), $l=2$ (d-subshell, 1), $m_l = -2, -1, 0, +1, +2$ (1 each, 5 values – pick one e.g., -2) and spin $+\frac{1}{2}$ or $-\frac{1}{2}$ (1). Total 5 marks. b) n determines radial distance – larger $n \rightarrow$ larger orbital (2). l determines shape and radial nodes – higher l (e.g., d) has more nodes, spreads electron density outward, increasing radius (2). Total 4. c) Both have same n, but $l=0$ (s) has no angular node and electron density is closer to nucleus, whereas $l=2$ (d) is more diffuse. Therefore X (s) will have smaller radius than Y (d); however shielding effects can make Y larger. Justify with radial distribution functions (2). Total=10.	[10]
46	a) Definition (1) + distinction covalent vs metallic (1)=2 marks. \nb) Correct sum ($152+112+85+77+71+66+57+54=734$) $\div 8 = 91.75$ pm (2 marks). Incorrect value (1) if arithmetic error, none (0). \nc) Correct explanation of Z_{eff} increase (2), shielding constant unchanged (1), radius decrease (1) =4 marks. Partial points for each concept. \nd) Correct description of graph: decreasing line, slope negative, confirms radius decrease across period (2). Partial for correct trend description (1). Total 10.	[10]
47	a) Correct application of Slater's rules: shielding 0.35 for each other 2p electron ($2 \times 0.35 = 0.70$) + 0.85 for 2s electrons ($2 \times 0.85 = 1.70$) + 1.00 for 1s electrons ($2 \times 1.00 = 2.00$). Total $S = 4.40$. $Z_{eff} = 7 - 4.40 = 2.60$ (4 marks). Partial for each shielding contribution (1-2 marks). \nb) Correct Z_{eff} for O (calculate similarly, result 3.55) and clear explanation that increased nuclear charge with similar shielding raises Z_{eff} (4 marks). Partial for correct value or qualitative trend (2-3 marks).	[8]

Q	MARKING POINTS (SPECIFIC)	MARKS
48	a) Correct identification of Al having a $3s^23p^1$ configuration and Si $3s^23p^2$; note that added p-electron in Si leads to increased effective nuclear charge and contraction (3 marks). Partial for mentioning electron repulsion or shielding (1-2 marks). \nb Correct prediction: radius continues to decrease from P to S; justification via increasing Z_{eff} and constant shielding (3 marks). Partial for correct trend only (1-2 marks).	[6]
49	a) Correct calculation: $((102-54)/102) \times 100 \approx 47\%$ (4 marks). Partial for correct formula (2) or correct value (2). \nb Correct explanation: same electron count (10) $\rightarrow$ same shielding, but nuclear charge rises from 11 to 13, increasing Z_{eff} and pulling electrons closer; also higher charge leads to stronger electron-electron repulsion reduction (4 marks). Partial for each concept (1-2 marks).	[8]
50	a) Correct link: bond length $\approx$ sum of two covalent radii; double bond involves greater s-character, reducing effective radius, leading to shorter bond (4 marks). Partial for mentioning covalent radius (2) and bond order effect (2). \nb Calculation: $2 \times 70 \text{ pm} = 140 \text{ pm}$; compare with observed 154 pm (difference +14 pm) and 134 pm (difference -6 pm). Discuss that additive model works reasonably for single bonds but overestimates double bond length due to increased s-character and pi-bond contraction (6 marks). Partial for correct arithmetic (2) and qualitative comment (2-4).	[10]